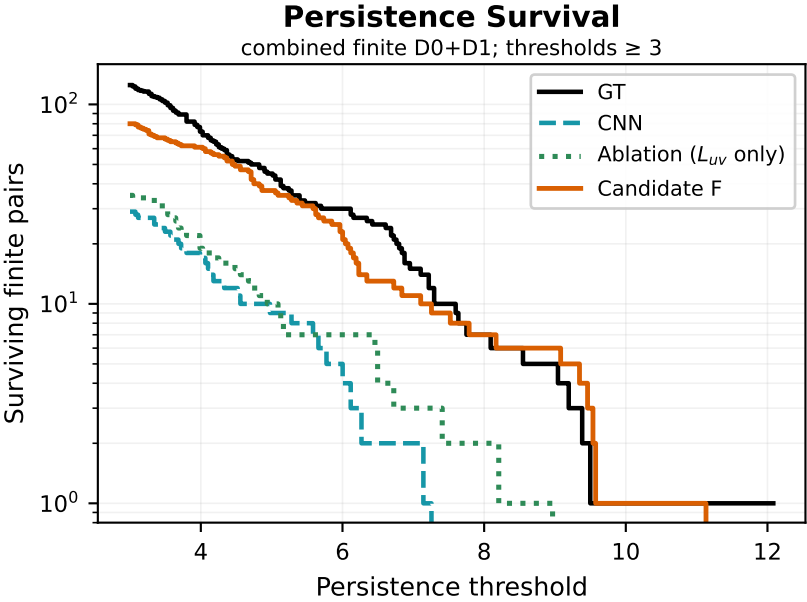
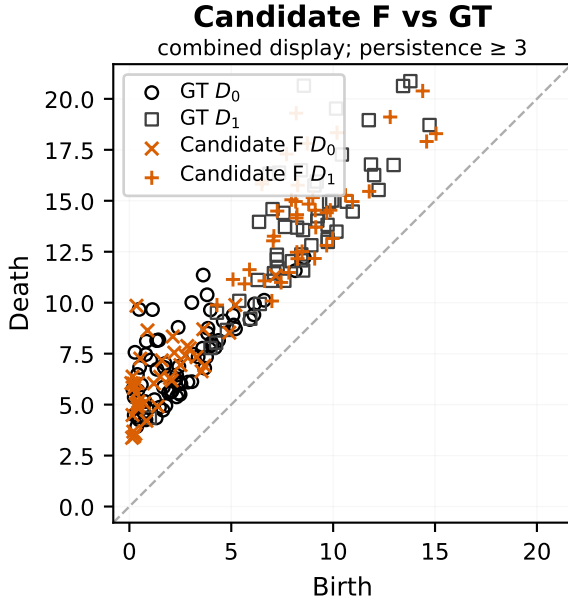
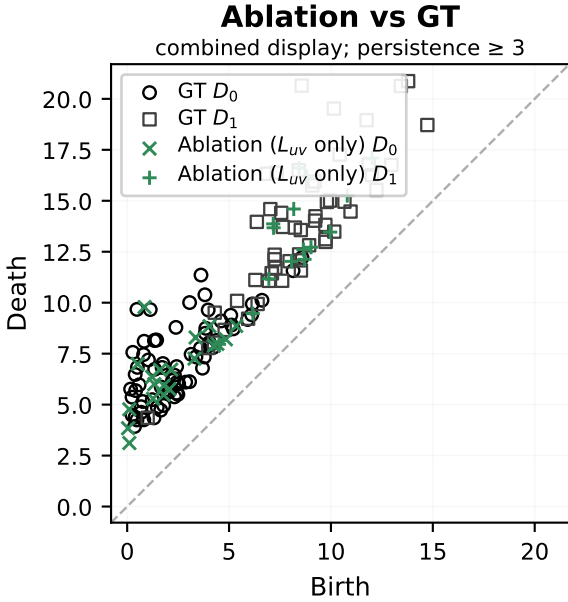
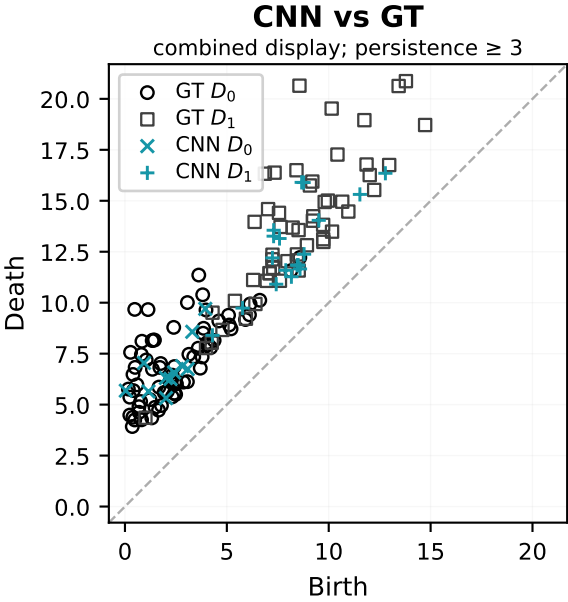
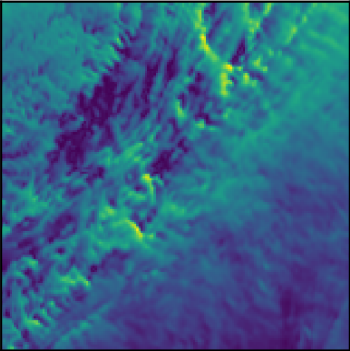


Sample 80: Topology-Aware Gains Beyond Matched Fine-Tuning

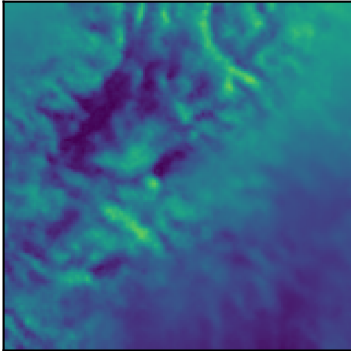
CNN = pretrained baseline; Ablation = matched L_{UV} -only fine-tuning; Candidate F = $L_{UV} + L_{grad}$ + repaired E2 supervision.



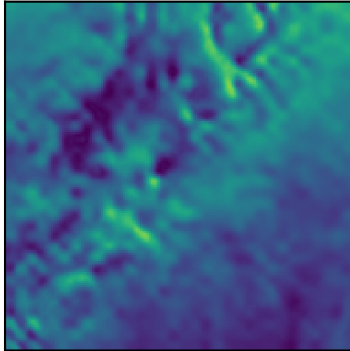
GT



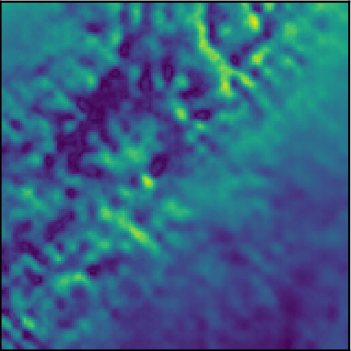
CNN



Ablation

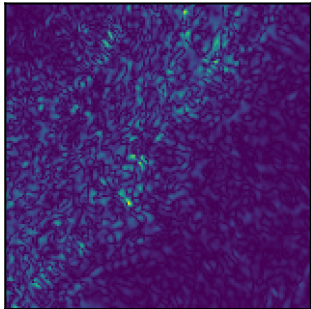


Candidate F

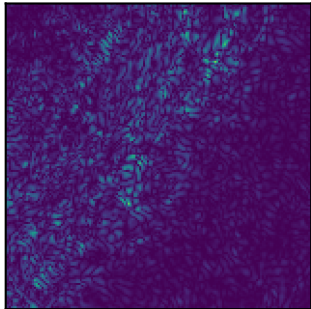


Wind-speed magnitude

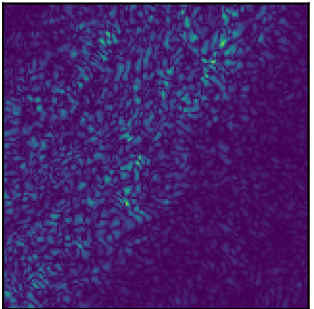
|CNN - GT|



|Ablation - GT|



|Candidate F - GT|



Absolute speed error

Topology-aware is better

$W_{2,\infty}$: CNN 22.770 → Abl. 23.272 → F 15.907
 $W_{2,2}$: CNN 29.627 → Abl. 30.484 → F 19.978

Conventional fidelity gaps

F vs CNN (original closeness rank 2/168): $|\Delta\text{PSNR}|=0.544$ dB, MAE=8.1%, RMSE=3.2%
F vs L_{UV} ablation: $|\Delta\text{PSNR}|=1.314$ dB, MAE=18.8%, RMSE=16.6%

Matched-control interpretation

Candidate F is lower than the matched L_{UV} -only fine-tuning control under all three validated PD metrics. This sample therefore supports an auxiliary-loss effect beyond ordinary fine-tuning.